

SHIELI, Kazakhstan

WMO: 380690

Lat: 44.182N

Lon: 66.729E

Elev: 502

StdP: 14.43

Time Zone: 5.00 (E05)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.7	0.8	-11.4	3.0	-5.3	-5.7	4.1	2.2	21.2	13.6	18.9	19.6	3.1	320	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.0	101.4	67.4	98.5	66.5	95.7	65.6	69.8	94.6	68.5	92.7	67.2	91.0	5.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	84.5	77.8	59.9	78.3	77.8	57.7	72.5	78.3	34.1	95.1	33.0	92.3	32.0	91.2	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	15.8	13.1	-8.6	106.1	9.3	2.4	-15.3	107.9	-20.7	109.3	-25.9	110.6	-32.7	112.4
			-9.1	72.5	9.1	2.1	-15.7	74.0	-21.0	75.2	-26.1	76.4	-32.7	78.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	25.1	29.0	45.1	57.7	69.9	77.9	80.3	77.5	66.4	52.5	38.7	27.2
	DBStd	21.70	11.02	12.05	9.74	9.34	6.75	6.05	5.58	6.20	7.32	8.63	9.81	10.13
	HDD50	2747	772	591	210	35	1	0	0	0	1	74	356	707
	HDD65	5575	1237	1009	618	251	32	2	0	2	69	393	790	1172
	CDD50	4231	0	2	59	266	617	837	938	852	492	152	16	0
	CDD65	1587	0	0	2	32	184	389	474	389	111	6	0	0
	CDH74	21416	0	0	37	468	2362	5317	6550	5041	1493	146	2	0
	CDH80	11310	0	0	8	143	1009	2923	3799	2780	624	24	0	0
Wind	WSAvg	4.8	4.2	5.0	5.6	5.9	5.9	4.7	4.6	5.1	4.7	4.1	3.6	4.1
Precipitation	PrecAvg	5.8	0.7	0.6	0.6	0.7	0.8	0.3	0.2	0.1	0.1	0.4	0.7	0.6
	PrecMax	10.9	1.9	1.9	1.6	3.4	3.3	1.2	1.6	0.7	0.7	1.3	2.6	1.6
	PrecMin	2.6	0.1	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
	PrecStd	2.0	0.4	0.4	0.4	0.6	0.8	0.4	0.4	0.2	0.2	0.4	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	63.1	78.8	90.2	96.4	104.1	106.4	103.5	96.2	83.2	71.2	52.8
		MCWB	44.9	49.7	56.6	61.7	63.1	68.6	68.7	69.2	64.8	57.9	54.3	43.6
	2%	DB	46.4	53.3	72.2	84.1	92.3	99.8	102.3	99.4	91.4	78.1	63.0	47.7
		MCWB	40.7	45.4	53.5	59.4	62.0	66.7	67.7	66.9	62.5	55.7	51.0	41.4
	5%	DB	41.6	48.3	66.1	79.2	89.2	96.9	99.0	96.5	87.4	73.6	57.9	44.2
		MCWB	37.2	41.5	51.1	57.5	61.5	65.9	67.0	65.9	61.0	54.3	47.8	39.4
	10%	DB	37.8	44.1	61.2	74.9	85.9	93.6	95.8	93.4	82.8	68.9	53.4	40.9
		MCWB	34.4	38.5	49.3	55.9	60.6	64.7	66.3	64.7	59.1	52.0	45.6	36.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	50.8	58.3	64.0	66.6	71.0	71.9	71.6	65.9	60.1	57.0	45.6
		MCDB	51.5	61.7	73.3	77.8	86.2	96.2	96.5	94.2	92.9	78.2	67.6	50.4
	2%	WB	41.3	45.9	55.0	61.2	64.7	69.0	70.0	69.3	63.5	57.1	52.4	42.3
		MCDB	45.8	52.2	69.3	78.6	85.2	93.6	94.4	94.2	88.2	75.0	62.0	46.5
	5%	WB	37.6	42.1	52.4	59.2	63.1	67.5	68.8	67.5	61.7	54.9	49.0	39.6
		MCDB	41.1	47.4	64.2	76.3	83.9	91.3	93.0	91.5	85.1	70.7	56.8	44.0
	10%	WB	34.6	38.6	49.9	57.2	61.7	66.2	67.5	65.9	59.7	52.6	45.8	36.8
		MCDB	37.7	44.0	60.5	73.5	82.0	89.7	91.4	89.6	81.1	67.9	52.9	40.8
Mean Daily Temperature Range	5% DB	MDBR	15.0	17.6	21.4	22.6	25.2	26.6	27.0	27.5	27.6	25.4	19.2	16.0
		MCDBR	17.5	22.0	28.6	29.3	29.7	30.7	30.6	30.9	32.0	31.5	26.1	19.0
		MCWBR	13.2	14.6	15.2	12.7	10.7	10.3	10.1	10.8	12.5	15.5	16.0	14.0
	5% WB	MCDBR	16.1	20.6	25.2	25.6	25.8	27.8	27.6	28.7	30.3	29.0	22.7	16.5
		MCWBR	12.6	14.3	14.5	11.9	10.5	10.5	10.1	11.3	12.6	15.2	14.5	12.8
Clear-Sky Solar Irradiance	taub	0.356	0.384	0.440	0.479	0.478	0.460	0.446	0.418	0.388	0.400	0.387	0.357	
	taud	2.090	2.053	1.988	1.930	1.976	2.068	2.128	2.185	2.217	2.103	2.104	2.107	
	Ebn at Noon	228	246	247	249	254	259	261	265	263	237	214	213	
	Edh at Noon	33	41	49	57	56	51	48	43	39	38	32	29	
All-Sky Solar Radiation	RadAvg	475	824	1259	1734	2212	2417	2374	2144	1687	1085	620	422	
	RadStd	69	87	98	140	131	107	99	72	62	67	62	52	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	-4.03	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	+1.67	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page